An approach to establishing the thresholds of plantar loading in obese children

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Introduction: Excessive weight-bearing positively affects the overloaded foot, which can induce multiple foot deformities [1]. Previous studies normalized maximum force to eliminate the influence of body weight on the mechanical loading of the foot [2]. To explore body weight itself to the change of the plantar pressure distribution, this study adopts a strategy of body weight scale to compare loading patterns between normal-weighted and obese children. It can acquire the exceeded foot loading data accurately for obese children compared to normal-weighted children, which could lead to finding the pressure threshold in obese children.

Research question: Is there a method to grade the pressure thresholds of plantar overload in obese children?

Methods: A cross-sectional study with a large sample size of 1170 participants aged 7-11 years was used to divide normal-weighted (n = 812) and obese children (n = 358) into eight groups based on the same weight class strategy: group 1 (25.5-30.4 kg), group 2 (30.5-35.4 kg), group 3 (35.5-40.4 kg), group 4 (40.5-45.4 kg), group 5 (45.5-50.4 kg), group 6 (50.5-55.4 kg), group 7 (55.5-60.4 kg), group 8 (60.5-65.4 kg). Dynamic plantar pressure data were collected using a Footscan® plantar pressure system (RSscan International, Belgium). Maximum forces were extracted from the main plantar region using principal component analysis.

Results: The change of obese children with the same weight grade compared with normal-weighted children was divided into six grades, to define the pressure threshold of obese children's plantar pressure compared with normal-weighted children. The assessment criteria of the pressure threshold level are set at 10 N (trivial effect), 10-20 N (very weak effect), 20-30 N (weak effect), 30-40 N (moderate effect), 40-50 N (strong effect) and 50-60 N (very strong effect). Table 1 shows the levels of the pressure threshold in obese children compared to normal-weighted children with the same weight class.

Group classification	Weight (kg)		Body mass index (kg/m²)		Maximum force variation (N)				Levels of pressure threshold			
	Normal-weighted children	Obese children	Normal-weighted children	Obese children	3rd MTPJ	4th MTPJ	Medial rearfoot	Lateral rearfoot	3rd MTPJ	4th MTPJ	Medial rearfoot	Lateral rearfoot
Group 1	27.9 ± 1.4	29.0 ± 0.2	15.6 ± 1.0	19.3 ± 0.2	-13.1	-12.6	3.0	-20.5	I	I	I	I
Group 2	33.0 ± 1.4	33.3 ± 1.6	16.6 ± 1.2	20.2 ± 0.9	-3.9	8.3	-30.8	-24.9	I	I	I	I
Group 3	38.0 ± 1.4	38.5 ± 1.4	17.5 ± 1.3	21.4 ± 1.1	17.3	11.3	-5.4	-7.7	II	II	I	I
Group 4	42.9 ± 1.5	43.2 ± 1.3	18.0 ± 1.1	22.8 ± 1.3	21.0	21.7	38.5	13.3	III	III	IV	II
Group 5	47.9 ± 1.4	48.1 ± 1.5	19.1 ± 1.2	23.8 ± 1.4	31.3	28.6	51.7	49.3	IV	III	VI	V
Group 6	52.5 ± 1.4	52.8 ± 1.5	19.7 ± 1.1	25.0 ± 1.4	14.2	34.5	41.2	33.7	II	III	V	IV
Group 7	56.6 ± 0.7	57.9 ± 1.4	19.9 ± 1.2	26.1 ± 1.7	41.5	36.7	48.4	26.8	V	IV	V	II
Group 8	62.6 ± 1.3	62.7 ± 1.4	21.0 ± 1.3	26.8 ± 1.8	45.1	47.8	48.5	57.0	V	V	V	VI

Discussion: The results showed that the maximum force of obese children with 25.5-35.4 kg did not cause significant damage to the main stress area of the plantar, and there was no need for clinical intervention or other related foot decompression strategies. When the weight of obese children is greater than 35.5 kg, it is necessary to pay attention to the influence of excessive foot load on the development and health of obese children's feet. This study can provide data support for foot decompression protocols such as shoes or insoles and weight loss training.

References

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A novel method for tracking movements of backpack's centre of mass in dynamic activities

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Introduction: Postural compensations with backpack may cause considerable body strains, resulting in fatigue, pain, and injury [1]. Backpack's position can influence carrier's posture and dynamic performance [2]. Characterizing the backpack's position, namely the position of its centre of mass (COM) with respect to the carrier's body, allows modelling its dynamic loading towards revealing the moment and moment of inertia it renders on the carrier. These knowledges will provide novel insights into the carrier's postural compensations and musculoskeletal injury [3]. Despite of the importance, there is a lack of an easy approach that can determine and track the movement of a backpack's COM during dynamic activities.

Research question: How to determine the position of a backpack's COM and track its movements in walking with the backpack?

Methods: A backpack was tightly filled with sandbags, resulting in a total weight of 10 kg. Using a 3D motion capture system (Vicon, UK), we created the backpack's local coordinate system (CS) with the three reflective markers attached on it. A directional cosine matrix was established for coordinate transformations between the backpack's and the lab's CS. A mannequin was then placed on an integrated force plate (Kistler, Switzerland), and its weight and centre of pressure were measured. This measurement was repeated after placing the backpack on the mannequin (two positions, Fig. 1a), and the horizontal coordinates of the backpack's COM were calculated according to the Varignon's Theorem.

Fig. 1. *Experiments and outcomes: a) Measuring centre of pressure in backpack's two postures; b) Displacement of backpack's and subject's COM during walking.*